

FREE BONUS GUIDE

Drill Press Mobile Base Plan

Build a sturdy rolling cabinet that fits most benchtop and floor-standing drill presses — with a full drawer for bits, clamps, and accessories.

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SECTION 1

Overview & Benefits

Why every workshop needs a mobile drill press station.

A drill press bolted to a bench wastes valuable workspace and limits where you can work. This mobile base puts your press on wheels with built-in storage.

What You'll Build

A rolling cabinet roughly 24" wide × 20" deep × 32" tall (61 × 51 × 81 cm) that supports most benchtop and floor-standing drill presses up to 250 lb (113 kg). A full-width drawer keeps drill bits, clamps, and sanding drums organized and within arm's reach.

Time Estimate

4–6 hours over a weekend, including finish drying time.

Skill Level

Beginner to Intermediate. Only basic joinery (butt joints reinforced with screws and glue).

Key Benefit

Roll your drill press to the workpiece instead of hauling heavy boards across the shop. When you're done, tuck it against the wall and reclaim floor space.

SECTION 2

Tools & Materials

Everything you need before you start cutting.

Tools Required

- Table saw or circular saw with a straight-edge guide
- Drill/driver with #2 Phillips and countersink bit
- Tape measure, combination square, pencil
- Clamps (at least 4 bar clamps, 24" / 61 cm)
- Random-orbit sander (120 and 220 grit)

Materials

ITEM	QTY	NOTES
3/4" (19 mm) plywood, 4×8 sheet	1	Birch or maple for strength
1/4" (6 mm) plywood	1 piece 22×18" (56×46 cm)	Drawer bottom
#8 × 1-1/4" (32 mm) wood screws	40	Coarse thread
Wood glue	8 oz	PVA, weatherproof not required
3" (76 mm) locking swivel casters	4	Rated 75 lb (34 kg) each minimum
18" (46 cm) full-extension drawer slides	1 pair	Ball-bearing, 100 lb (45 kg) rated
Drawer pull	1	4–5" (10–13 cm) bar style
Polyurethane or lacquer	1 can	Spray or brush-on

SECTION 3

Cutting List

All parts from one 4×8 sheet of 3/4" plywood.

PART	QTY	WIDTH	LENGTH	THICKNESS
Top	1	20" (51 cm)	24" (61 cm)	3/4" (19 mm)
Sides	2	20" (51 cm)	28" (71 cm)	3/4" (19 mm)
Back	1	22-1/2" (57 cm)	28" (71 cm)	3/4" (19 mm)
Bottom shelf	1	20" (51 cm)	22-1/2" (57 cm)	3/4" (19 mm)
Drawer front	1	5" (13 cm)	21" (53 cm)	3/4" (19 mm)
Drawer sides	2	5" (13 cm)	18" (46 cm)	3/4" (19 mm)
Drawer back	1	4-1/4" (11 cm)	19-1/2" (50 cm)	3/4" (19 mm)
Drawer bottom	1	18" (46 cm)	20-1/2" (52 cm)	1/4" (6 mm)

Tip

Label each piece with painter's tape right after cutting. It saves confusion during assembly, especially with the similarly sized sides and back panels.

SECTION 4

Building the Frame

Assemble the cabinet box step by step.

- 1 Prepare the sides.** Sand both faces of each side panel to 120 grit. Mark the inside face and draw a line $3/4"$ (19 mm) from the bottom edge — this is where the bottom shelf sits.
- 2 Attach the bottom shelf.** Apply glue along both ends of the shelf. Position it on the marked lines and clamp. Pre-drill and drive 4 screws per side, spaced evenly.
- 3 Attach the back panel.** Stand the assembly upright. Apply glue to the rear edges of the sides and bottom shelf. Align the back panel flush with the top and outside edges. Clamp and screw every 6" (15 cm).
- 4 Attach the top.** Apply glue to the top edges of the sides and back. Set the top panel with equal overhang front and sides. Clamp and screw from underneath through the sides (4 screws per side) and through the back (3 screws).
- 5 Check for square.** Measure both diagonals — they should be within $1/8"$ (3 mm). Adjust before the glue sets.

SECTION 5

Adding the Drawer

A full-width drawer for organized tool access.

- 1 Build the drawer box.** Glue and screw the drawer sides to the drawer front and back, forming a rectangular box. Slide the 1/4" (6 mm) plywood bottom into the grooves or simply glue and pin-nail it to the underside.
- 2 Mount the slides.** Mark a line 8" (20 cm) from the bottom of each cabinet side. Align the cabinet half of each slide on this line and screw in place. Attach the drawer half to each drawer side, flush with the bottom edge.
- 3 Test the fit.** Slide the drawer in. It should glide smoothly with no binding. Adjust the slides if needed before continuing.
- 4 Attach the drawer pull.** Center the pull on the drawer front and drill mounting holes. Bolt from inside the drawer.

Tip

If you want dividers inside the drawer, cut strips of 1/4" plywood to height and slot them in. No glue needed — friction fit lets you rearrange later.

SECTION 6

Installing Casters

Make it roll — and stop when you need it to.

- 1 Mark caster positions.** Flip the cabinet upside down. Mark each caster position 1" (2.5 cm) in from each corner of the bottom shelf.
- 2 Pre-drill.** Use a bit slightly smaller than the caster mounting screws. Drill through the bottom shelf only.
- 3 Bolt the casters.** Use the bolts included with your casters. Add a flat washer on top for load distribution. Tighten firmly.
- 4 Test the locks.** Flip the cabinet upright and test each caster lock. The cabinet should not move when all four are engaged.

Important

Always lock all four casters before operating the drill press. A rolling base under a spinning bit is a safety hazard.

SECTION 7

Finishing & Final Assembly

Protect the wood and mount your drill press.

- 1 Sand to 220 grit.** Go over every visible surface. Round sharp edges slightly for a professional feel.
- 2 Apply finish.** Two coats of polyurethane or lacquer, sanding lightly with 320 grit between coats. Let dry fully (follow the can's instructions).
- 3 Mount the drill press.** Center the press on the top panel. Mark the mounting bolt holes through the press's base. Drill through the top and bolt from underneath using carriage bolts, washers, and lock nuts.
- 4 Organize the drawer.** Add your drill bits, chuck key, clamps, and sanding drums. Everything within arm's reach while you work.

Final Checklist

- ☐ All screws tight and countersunk
- ☐ Drawer slides smoothly with no binding
- ☐ All four casters lock securely
- ☐ Drill press bolted down and does not rock
- ☐ Finish fully cured before first use

Next Step

Your mobile drill press base is ready. Roll it into position, lock the casters, and start drilling.

For more workshop organization projects, open **The Ultimate Woodworking Plans Vault** and browse the *Workshop* category — you'll find plans for tool cabinets, lumber racks, clamp storage, and more.